SAT MATH FORMULAS SHEET

Linear equations

Slope-intercept form	$y = mx + b$ where slope = m y-intercept = b
Point-slope form	$y - y_1 = m\,(x - x_1)$ where slope = m point on line (x_1, y_1)
Standard form	$ax + by = c$ where slope = - a/b y-intercept = c/b x-intercept = c/a
Slope	$m = \frac{y_2 - y_1}{x_2 - x_1}$

Exponents / Radicals

$x^a x^b$	x^{a+b}
$\frac{x^a}{x^b}$	x^{a-b}
$(x^a)^b$	x^{ab}
$(xy)^a$	$x^a y^a$
x^{-a}	$\frac{1}{x^a}$
x^0	$1 \ (if\, x \neq 0)$
$x^{\frac{a}{b}}$	$\sqrt[b]{x^a}$
$\sqrt{xy}$	$\sqrt{x}\sqrt{y}$
$\sqrt{\frac{x}{y}}$	$\frac{\sqrt{x}}{\sqrt{y}}$

Trigonometry

$sin\ \alpha$	Opposite/Hypothenuse
$cos\ \alpha$	Adjacent/Hypothenuse
$tan\ \alpha$	Opposite/Adjacent
$sin(90 - \alpha)$	$cos\ \alpha$
$cos(90 - \alpha)$	$sin\ \alpha$
$sin^2\alpha + cos^2\alpha$	1
30^0 to radians	$30^0 = \frac{\pi}{6}$ radians
45^0 to radians	$45^0 = \frac{\pi}{4}$ radians
Common Pythagorean triples	3-4-5 6-8-10 5-12-13

Quadratics

Factored Form	$y = a\,(x - p)(x - q)$
x-intercepts	$x = p$ or $x = q$
Vertex	$x = (p + q)/2$
Vertex Form	$y = a\,(x - h)^2 + k$
Vertex	(h, k)
Standard Form	$y = ax^2 + bx + c$
Vertex	$x = \frac{-b}{2a}$
x-intercepts	1) Try factoring 2) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
When are there 2 real x-intercepts?	$b^2 - 4ac > 0$
When is there 1 real x-intercept?	$b^2 - 4ac = 0$
When are there no real x-intercepts?	$b^2 - 4ac < 0$

Miscellaneous

Complex number	$i^2 = -1$ or $i = \sqrt{-1}$
Average or arithmetic mean	sum / number of values
Relationship of distance, rate, and time	$d = rt$ where distance = d rate = r time = t
Standard Form of the equation of circle	$(x - h)^2 + (y - k)^2 = r^2$ where (h, k) center of the circle and r is radius
Area of circle	$A = \pi r^2$
Circumference of circle	$C = 2\pi r$
Arc length	$\frac{\alpha}{360} 2\pi r$
Sector area	$\frac{\alpha}{360} \pi r^2$
Distance	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$